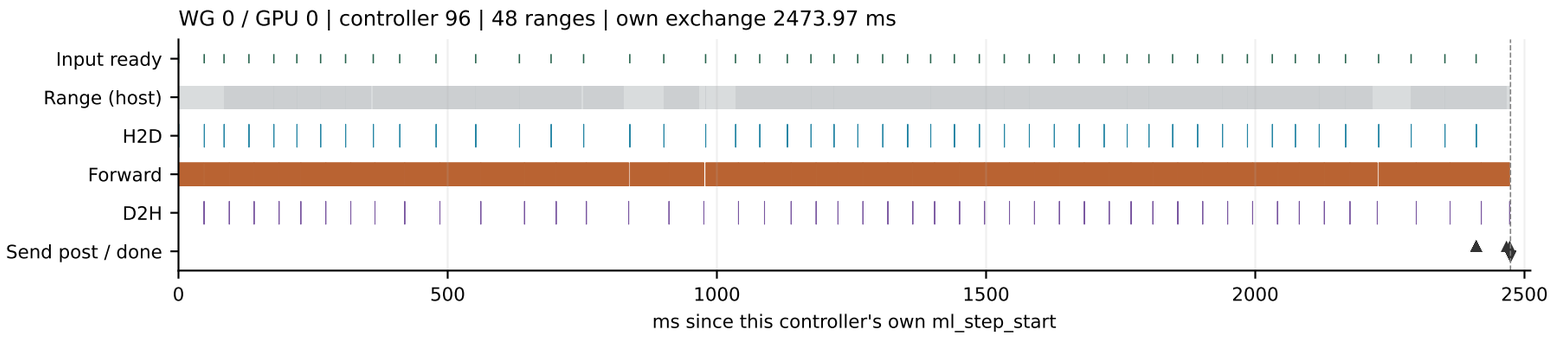
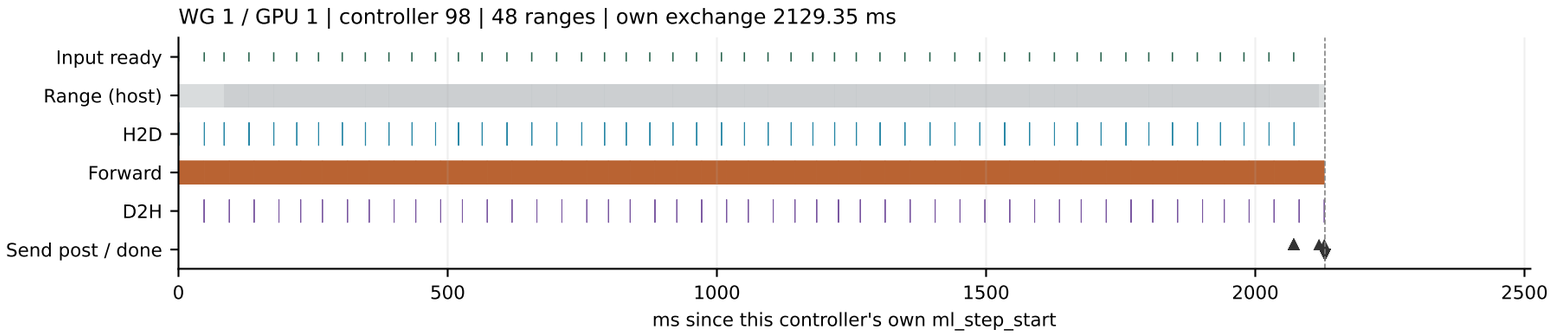
transformer | 96p96c4g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 1748.16 ms | p2p1 2487.44 ms



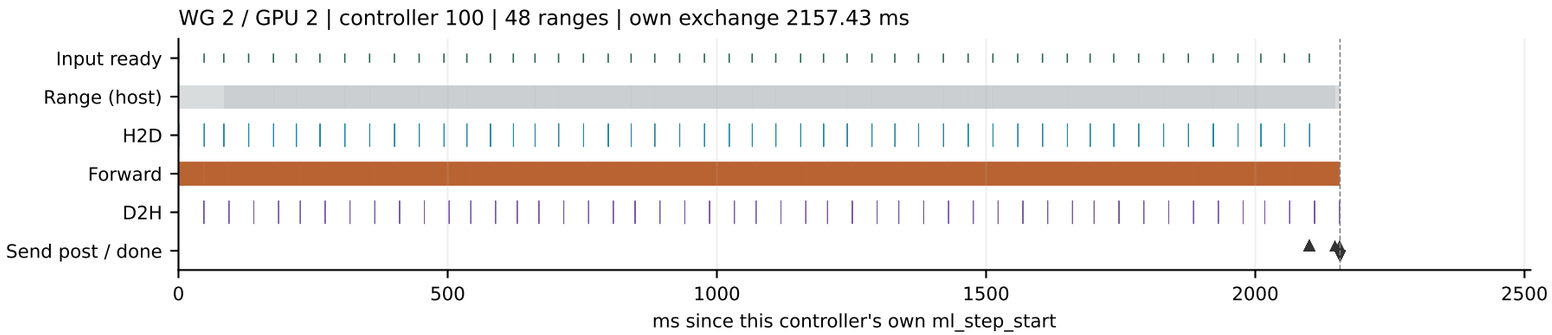
GPU work (ms)	Collective	p2p1
H2D	5.401	15.132
Forward	1665.057	2461.530
D2H	7.200	7.564
GPU SUM	1677.658	2484.226

Controllers: coll 96 / P2P 96
Coll host device elapsed: 1678.708 ms
P2P GPU envelope: 2472.352 ms
Different scopes; not an elapsed-time comparison.



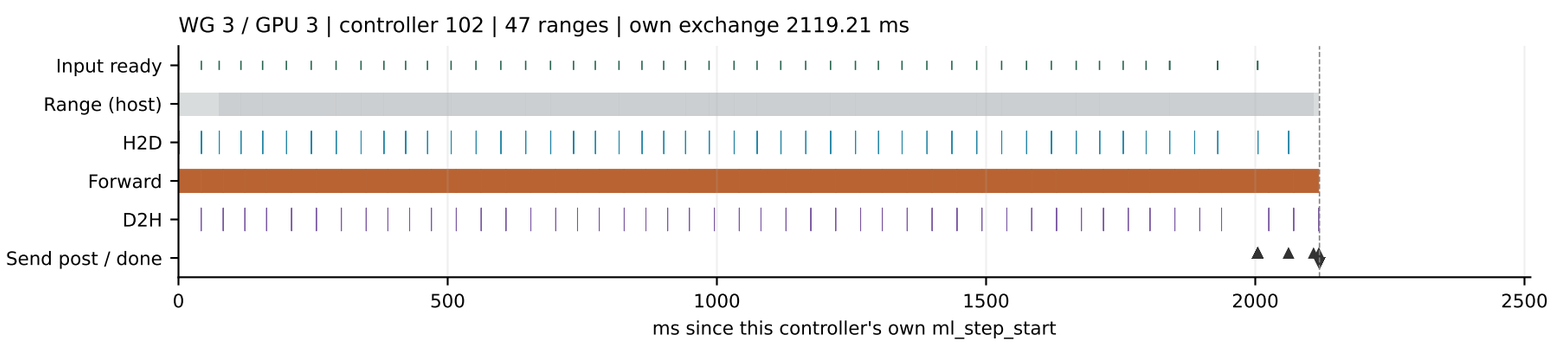
GPU work (ms)	Collective	p2p1
H2D	5.779	17.277
Forward	1617.951	2126.282
D2H	6.283	7.369
GPU SUM	1630.013	2150.927

Controllers: coll 98 / P2P 98
Coll host device elapsed: 1630.566 ms
P2P GPU envelope: 2128.004 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	5.097	15.591
Forward	1657.789	2154.642
D2H	5.643	7.533
GPU SUM	1668.528	2177.766

Controllers: coll 100 / P2P 100
Coll host device elapsed: 1669.044 ms
P2P GPU envelope: 2156.232 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	5.159	18.472
Forward	1607.614	2115.709
D2H	4.607	7.325
GPU SUM	1617.380	2141.505

Controllers: coll 102 / P2P 102
Coll host device elapsed: 1618.148 ms
P2P GPU envelope: 2117.524 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.